



I am attaching the results showing the number of NAs for key variables for all countries that contain NA values. I also attached the correlation matrix for these key variables (omitting NA values).

Based on concerns about attribution bias and multicollinearity, what would you suggest I do? Is it reasonable and academically justified to remove `log_gdp` here? could electricity access sufficiently proxy for statistical capacity (ability) in the same way that `log_gdp` was supposed to? What about luminosity? What if there isn't high enough correlation (refer to screenshots)? Please help me figure this out.

Short answer: yes, it is reasonable and defensible to drop log GDP from your main TWFE, and rely on electricity access (and possibly luminosity) as development/state-capacity controls, as long as you (1) explain why and (2) show robustness checks that include GDP on the reduced sample.

What the two screenshots are telling you

From the missingness table:

- Almost all remaining NAs are in `log_gdppc`, plus a smaller number in `luminosity`, with essentially none in `electric access`.
- And those NAs are concentrated in exactly the kinds of countries you care about (North Korea, Eritrea, Syria, Venezuela, etc.), which are highly autocratic.

From the correlation matrix:

- `log_gdppc` is strongly correlated with:
 - `electric_access` (~0.71),
 - your social progress index `spi_overall` (~0.62),
 - and `lib_dem` (~0.61).
- `luminosity` is moderately correlated with `log_pop` (~0.45) and weak-to-moderately with `log_gdppc` (~0.23) and `electric_access` (~0.19).

So GDP is both:

- A classic development proxy, and
- Highly collinear with other development/capacity proxies you already have.

Attribution bias vs multicollinearity

1. Attribution / selection bias

Including log GDP forces you to drop 411 country-years, and those are disproportionately in exactly the regimes where data is politically suppressed. That means:

- Your treatment (`aut_ep / lib_dem`) is correlated with whether GDP is observed.
- If you condition on “GDP available and non-NA,” you’re selecting on a post-treatment variable (data quality that is itself shaped by politics).
- That pushes your sample toward relatively well-governed, transparent countries and makes your estimates too conservative about the true effect of autocratization on missingness.

So from an identification perspective, keeping `log_gdppc` may hurt you more than it helps.

2. Multicollinearity

Given those correlations, log GDP, electric access, and SPI are all measuring overlapping “development / capacity” space. That:

- Doesn’t bias coefficients, but
- Inflates standard errors and makes it harder to separately identify each one.

Your correlation matrix is in the “mild to moderate collinearity” zone: it’s not catastrophic, but it’s clear you don’t *need* all three in the same equation to get a reasonable adjustment for development.

Is it academically OK to drop `log_gdppc`?

Yes, especially in this paper, where:

- GDP is missing non-randomly in regimes of interest.
- You have theoretical reasons to treat GDP availability itself as politically driven.
- You retain other plausible controls for development and capacity.

A clean, defensible strategy:

1. Main specification (preferred):

- Drop `log_gdppc`.
- Keep `log_pop`, `electric_access`, `rural_pop_pct`, and (if you like) `luminosity`.
- Interpret these as proxies for size, infrastructure, and state capacity.

2. Robustness checks:

- Add `log_gdppc` back in on the reduced sample where it’s observed and re-run the models.

- Show that the sign and magnitude of the democracy/autocratization coefficient are similar, even if SEs get larger.
- Optionally, run a model with `log_gdppc` only (dropping electric access & luminosity) to show results don't hinge on which development proxy you use.

Write this explicitly in the paper: GDP is dropped in the main model *because* its missingness is endogenous to the political processes you study, and because other proxies capture development without inducing as much selection.

Can electricity access “stand in” for GDP?

Yes, as a proxy for development/state capacity, it's actually more interpretable for your question:

- It captures physical infrastructure, service provision, and the state's ability to deliver basic utilities.
- It is highly correlated with GDP and SPI in your data, meaning it tracks much of the same development variation, but with far fewer missing values.
- It's also substantively aligned with your story: states that cannot/will not maintain infrastructure or collect/utilize statistics tend to have low electrification.

So using electricity access (+ maybe luminosity) as your main development/capacity controls is reasonable and easy to justify.

What about luminosity and “not enough correlation”?

You don't need luminosity to be extremely correlated with GDP to justify including it:

- Luminosity captures economic activity and infrastructure at night; it often tracks informal or under-reported activity that GDP misses.
- The moderate correlations you see suggest it adds complementary information, not pure redundancy.
- The only time you'd worry is if correlations were too high (e.g., >0.9) and VIFs blew up. “Not high enough” is not a problem.

Given your outputs, I'd treat luminosity as an *optional extra*:

- Include it in the main model if it doesn't cause big VIFs.
- If it does, you can drop it and keep just `electric_access` as your capacity proxy.

Concrete recommendation

For the paper, I'd do this:

1. Main TWFE model

- Outcome: SDG missingness (goal-level and/or overall).

- Key predictor: `lib_dem` (continuous) or `aut_ep` (binary).
- Controls: `log_pop`, `electric_access`, `rural_pop_pct` (and optionally `luminosity`).
- Fixed effects: `country` + `year`.

2. Robustness section

- Add `log_gdppc` on the restricted sample; compare coefficients.
- Briefly describe which countries are excluded when GDP is required (using the NA table you just built).

3. Textual justification

- Argue that GDP is conceptually related to your outcome and treatment *and* missing not at random, so including it as a control would “control away” part of the political process you want to study while shrinking your sample in a highly non-random way.
- Emphasize that retaining electric access (and maybe luminosity) keeps a reasonable adjustment for development and capacity without those drawbacks.

That combination is methodologically sound and very easy to defend to reviewers.

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1. Screenshot2026-05-24-at-8.00.40-PM.jpg

2. Screenshot2026-05-24-at-8.00.52-PM-2.jpg